

Figure 1: STUN, TURN and signalling in WebRTC

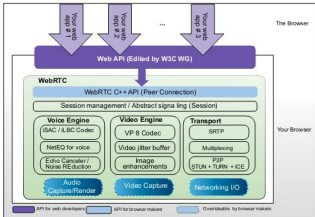


Figure 2: WebRTC architecture

as a fallback, relaying data between end points. TURN is used to relay audio, video and data streaming between peers and is not used for signalling data. TURN servers have public addresses, so they can be contacted by peers even if the peers are behind firewalls or proxies. TURN servers have a conceptually simple task — to relay a stream. For a WebRTC call, once the capabilities of the browser or application are exchanged, a point-to-point connection is established. This includes NAT traversal.

Features of WebRTC

The objective of the WebRTC project is to enable rich, high quality, RTC applications to be developed for the browser, mobile platforms, and IoT devices, and allow them all to communicate via a common set of protocols.

- WebRTC allows Web platforms to include audio conferencing, video conferencing and data sharing as part of the websites or applications
- It allows a Web browser as an end point for a call
- It allows websites access to video camera feeds to manipulate the feed or to allow users to record video
- It provides platform and device independence so that developers can write WebRTC HTML 5 code that will run across different OS browsers

- Advanced audio and video quality is provided by using the Opus audio codec and the VP8 video codec

WebRTC architecture

There are many components to the WebRTC architecture that readers need to familiarise themselves with. The most important are listed below.

Web API: The Web API is used to create Web applications, which may have video and audio chat capabilities. The Web API is used by third party developers to develop Web based video chat-like applications.

WebRTC Native C++ API: The WebRTC Native C++ API layer enables browser developers to easily implement the Web API proposal. This is used by browser developers to create WebRTC compatible browsers, and they can override the audio capture, video capture and the network I/O render.

Voice engine: This sets the framework for the audio media connection, which includes the iSAC (Internet Speech Audio Codec), the iLBC (Internet Low Bitrate Codec), the Opus codec, the NetEQ for Voice, the acoustic echo canceller and noise reduction technologies. The iSAC / iLBC / Opus audio codec for VoIP and audio streaming over wideband and narrowband supports constant and variable bitrate encoding from 6 kbits/s to 510 kbits/s, as well as frame sizes from 2.5 ms to 60 ms.

The NetEQ for Voice is a dynamic jitter buffer and error concealment algorithm used for concealing the negative effects of network jitter and packet loss, which keeps latency as low as possible while maintaining the highest voice quality.

The acoustic echo canceller is a software based signal processing component that removes the acoustic echo resulting from the voice being played out coming into the active microphone. The noise reduction component is a software based signal processing component that removes certain types of background noise usually associated with VoIP, such as a hiss, fan noises and other background disturbances.

Video engine: This is the framework for the video media connection for video, from the camera to the network, and from the network to the screen and back. It includes the VP8 (video compression format owned by Google) codec, dynamic jitter buffer and image enhancement technology. The VP8 codec by the WebM Project (an open media file format designed for the Web) is well suited for RTC as it is designed for low latency. The dynamic jitter buffer for video helps conceal the effects of jitter and packet loss on overall video quality, and image enhancements remove video noise from the image capture by the webcam.

Transport session: Transport session components are built by re-using components from libjingle, without using or requiring the Extensible Messaging and Presence Protocol (XMPP) or Jingle Protocol. The Real Time Protocols (RTP) network stack and the STUN and ICE components allow calls to establish connections across various types of networks.